2. **Osteoporosis:** It occurs due to deficiency of vitamin D and hormonal imbalance. The bone becomes soft and fragile. It causes rickets in children and osteomalacia in adult females. It can be minimized with adequate calcium intake, vitamin D intake and regular physical activities.

9.12 Benefits of regular Exercise

Exercise and physical activity fall into four basic categories. Endurance, Strength, Balance and Flexibility.

Endurance or aerobic activities increase the breathing and heart rate. They keep the circulatory system healthy and improve overall fitness.

Strength exercises make the muscles stronger. They help to stay independent and carry out everyday activities such as climbing stairs and carrying bags.

Balance exercises help to prevent falls which is a common problem in older adults. Many strengthening exercises also improves balance.

Flexibility exercises help to stretch body muscles for more freedom of joint movements. Regular exercises can produce the following beneficial physiological changes:

- The muscles used in exercise grow larger and stronger.
- The resting heart rate goes down.
- More enzymes are synthesized in the muscle fibre.
- Ligaments and tendons become stronger.
- Joints become more flexible.
- Protection from heart attack.
- Influences hormonal activity.
- Improves cognitive functions.
- Prevents Obesity.
- Promotes confidence, esteem.
- Aesthetically better with good physique.
- Over all well-being with good quality of life.
- Prevents depression, stress and anxiety.

During muscular exercise, there is an increase in metabolism. The O_2 need of the muscles is increased. This requirement is met with more oxygen rich RBCs available to the active sites. There is an increase in heart rate and cardiac output. Along with balanced diet, physical activity plays a significant role in strengthening the muscles and bones.

Summary

Movement is one of the significant features of living organisms. The different types of movements are amoeboid movement, ciliary movement, flagellar movement and muscular movement. Three types of muscles are present in human beings. They are the skeletal muscle, visceral muscle and cardiac muscle. The skeletal muscles are attached to the bones by tendons.

The most striking microscopic feature of skeletal muscle is a series of light and dark bands. The muscles exhibit the properties such as excitability, contractibility, conductibility and elasticity. There are two types of muscle contraction. They are isotonic and isometric contractions.

The skeletal system consists of a frame work of bones and cartilages. The skeletal system is grouped into two principal divisions: the axial skeleton and the appendicular skeleton. There are three types of joints present in the body: fibrous, cartilaginous and synovial joints.

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